



MARMARA
ÜNİVERSİTESİ

Introduction to Databases

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Database System Project - 1. Phase

Şerefim ve inandığım tüm değerler adına yemin ederim ki bu projede hiçbir yerden kopya

çekmedim ve copy paste yapmadım.

I swear, in the name of my honor and all the values I believed in, that I did not copy and paste

anything from anywhere during this project.



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Brief Description of the System

In this project, we designed and implemented a Bicycle Rental Database System that manages all the operational activities of a bike rental company.

The main goal of the system is to provide a well-organized, secure, and automated structure where bicycles, customers, reservations, insurance policies, penalties, and maintenance operations can be efficiently tracked and controlled.

In modern cities, bicycle rental services play an important role in sustainable urban transportation. Managing these operations manually often results in data redundancy, inconsistent records, and inefficient tracking of maintenance or insurance dates.

Therefore, our team developed a relational database system that automates the following key operations:

- registering customers and bicycles,
- handling reservation and payment processes,
- calculating rental costs based on class and duration,
- recording maintenance activities performed by staff members,
- and ensuring that each bicycle is covered by a valid insurance policy.

The system provides instant data access for authorized users and ensures data integrity through primary and foreign key relationships. It aims to improve the accuracy of financial transactions, reduce human errors, and enhance the overall efficiency of the bicycle rental process.

1.1 Scope Definition

The scope of our project covers all database operations required to manage the daily workflow of a bicycle rental company — from customer registration and reservation creation to maintenance tracking and insurance validation.

Our system allows:

- Customers can make reservations for available bicycles by using the unique QR code assigned to each bike.
- Employees to track reservation and penalty.
- Maintenance personnel to log maintenance details, update service status, and plan upcoming maintenance dates.

All information is securely stored in a centralized relational database, accessible only to authorized users such as administrators, and maintenance personnel. The system ensures data consistency, accuracy, and efficiency in every operational process.

1.3 Determining System Requirements

Functional Requirements:

- The system must store and manage customer information such as full name, phone number, and email.
- Each bicycle must have a unique QR code, status, and related model, class, and insurance information.
- The system must allow customers to make reservations only for bicycles that are available.
- Each reservation must automatically calculate the total payment based on bicycle class and rental duration.
- The system must record penalties for late returns or damaged bicycles.
- Maintenance personnel must be able to log maintenance operations and future service dates.
- All payments and financial transactions must be securely stored and linked to the related reservations.
- Authorized users must be able to view and update data according to their role (customer, maintenance personnel).

1.3 Entities and Descriptions

A) Customer

The customer is the individual who rents a bicycle through the system. Each customer must be registered in the database with a unique customer ID.

The Customer entity stores essential personal information such as full name, phone number, and email address. Each customer can make multiple reservations and may have several related payments and penalties recorded in the system.

B) Bicycle Class

This entity defines the pricing structure and main categories of bicycles. Each class is identified by a unique bicycle class ID, which helps organize bicycles by their type — such as mountain, city, or electric bikes. The entity includes pricing details such as price per day and price per minute, which determine the total rental cost according to the class and duration of use. Each bicycle class has a distinct class name and can be linked to multiple bicycle models.

C) Bicycle Model

The Bicycle Model entity describes the technical specifications and brand information of bicycles. Each model is identified by a unique model ID and is associated with a single bicycle class ID . The entity stores attributes such as bicycle brand, seat capacity, and wheel count. Each model belongs to one class but can represent multiple bicycles in the system.

D) Bicycle

The Bicycle entity represents each physical bicycle available for rent. Every bicycle has a unique QR code used for digital identification and tracking. It also includes attributes such as status, model ID, and insurance policy number, which define the current condition, model relationship, and insurance coverage of the bicycle. Each bicycle can be reserved multiple times, but belongs to only one model and one insurance policy.

E) Insurance

The Insurance entity stores insurance details for each bicycle. It includes fields such as policy number, insurance name, begin date, expiry date, and price per year. Each insurance policy ensures financial protection for bicycles against damage or accidents during active reservations. One insurance record can be linked to multiple bicycles if they share the same policy.

F) Reservation

The Reservation entity records the details of the rental transaction between a customer and a bicycle. Each reservation has a unique reservation ID and contains attributes such as start time, end time, payment, payment method, and time lag. Each reservation is connected to one customer and one bicycle, and during the reservation period, the bicycle's status changes to "rented." After completion, the system automatically generates payment or penalty records if necessary.

G) Penalty

The Penalty entity contains records of additional charges imposed on customers for rule violations, such as late returns or damaged bicycles.
Each penalty has a unique penalty ID, an amount, and a reason.
Every penalty record is linked to one specific reservation, ensuring financial accountability and accurate billing.

H) Maintain

The Maintain entity stores maintenance and repair information for bicycles.
It includes attributes such as maintenance date, next maintenance date, and work description.
Each maintenance record is linked to a specific bicycle and a maintenance staff member.
This entity ensures that bicycles remain safe, functional, and ready for future rentals.

I) Maintain Personal

The Maintain Personal entity represents the employees who perform or record maintenance activities.
Each staff member is identified by a unique staff ID and has attributes such as full name and role (e.g., technician, manager, staff).
One maintenance staff member can perform multiple maintenance operations and is responsible for updating maintenance details in the database.

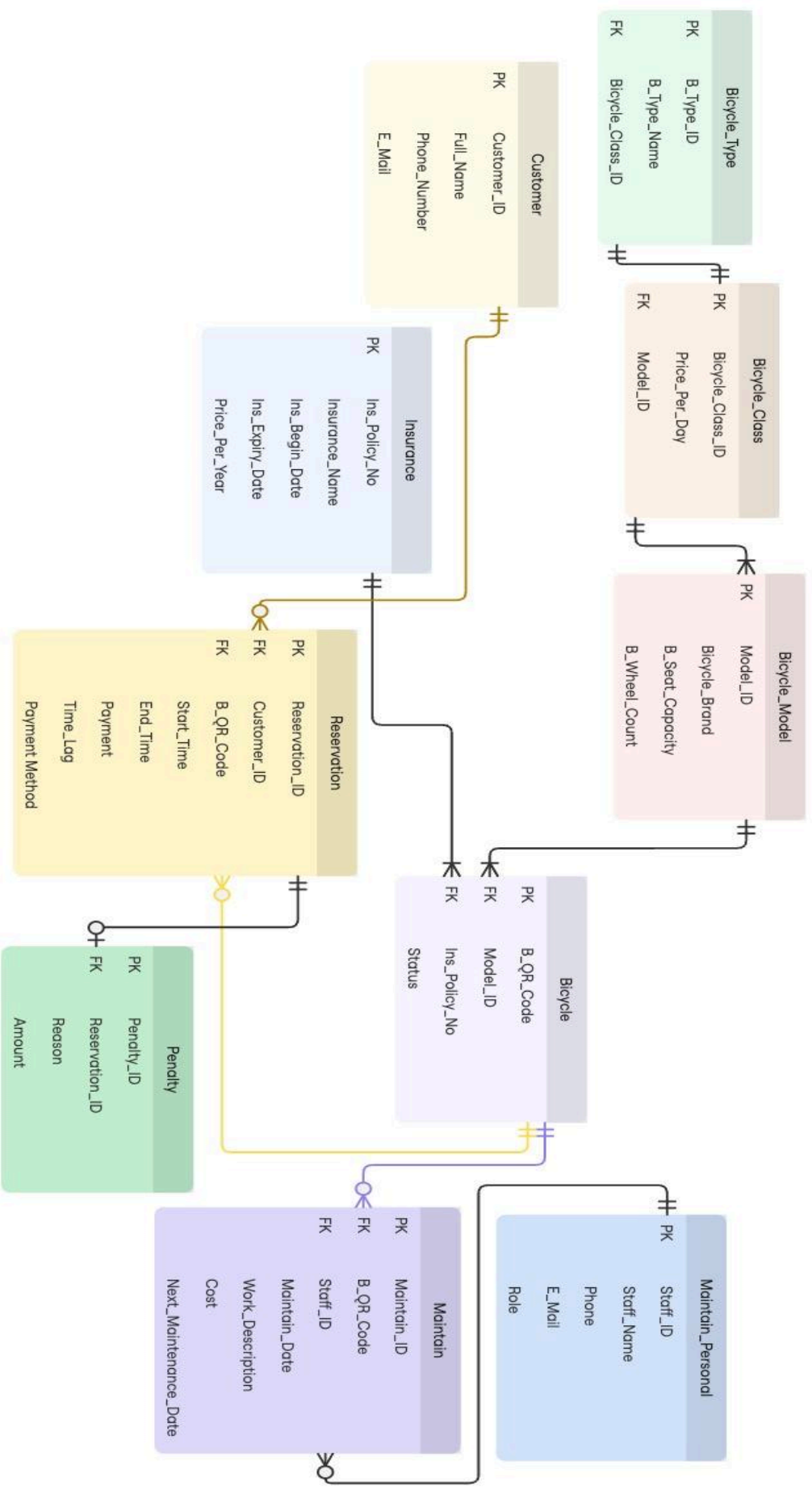
Data Dictionary

Table Name	Attribute Name	Description	Type	PK/FK	Required	Allowed Values / Constraint
Customer	Customer_ID	Unique customer identifier (PK).	INT	PK	Y	Range: 1 – 2,147,483,647
	Full_Name	Customer's full name.	VARCHAR(100)		Y	Alphabetic characters (A–Z, a–z), space, hyphen. Example: "Ahmet Muhtar"
	Phone_Number	Customer's phone number.	VARCHAR(13)		Y	Must follow +country/area formatting (e.g., +905XX...).
	E_Mail	Customer's email address.	VARCHAR(100)		Y	Must contain "@" and a valid domain (e.g., user@example.com).
Bicycle_Class	Bicycle_Class_ID	Unique bicycle class identifier	TINYINT	PK	Y	Range: 1 – 255
	Price_Per_Day	Per-day price for rentals.	DECIMAL(6, 2)		Y	Range: 0.00 – 999.99
	Model_ID	Foreign Key to the bicycle model.	SMALLINT	FK (Bicycle_Model)	Y	Must reference a valid Bicycle_Model(Model_ID) record.
Bicycle_Type	B_Type_ID	Unique bicycle type identifier.	TINYINT	PK	Y	Range: 1 – 255
	B_Type_Name	Name of the bicycle type.	VARCHAR(30)		Y	Allowed values: 'Standard', 'Electric', 'Mountain', 'Road', 'Kids', 'Tandem'.
	Bicycle_Class_ID	Foreign Key to the bicycle class.	TINYINT	FK (Bicycle_Class)	Y	Must reference a valid Bicycle_Class(Bicycle_Class_ID) record.
Bicycle_Model	Model_ID	Unique bicycle model	SMALLINT	PK	Y	Range: 1 – 32,767

		identifier.				
	Bicycle_Brand	Brand of the bicycle.	VARCHAR(50)		Y	Alphanumeric text.
	B_Seat_Capacity	Bicycle's seat capacity.	TINYINT		Y	Range: 1 or 2
	B_Wheel_Count	Bicycle's wheel count.	TINYINT		Y	Range: 2 or 3
Bicycle	B_QR_Code	Unique identifier/QR code for the bicycle.	VARCHAR(10)	PK	Y	Unique alphanumeric code.
	Model_ID	Foreign Key to the bicycle model.	SMALLINT	FK (Bicycle_Model)	Y	Must reference a valid Bicycle_Model(Model_ID) record.
	Ins_Policy_No	Foreign Key to the insurance policy number.	VARCHAR(20)	FK (Insurance)	Y	Must reference a valid Insurance(Ins_Policy_No) record.
	Status	Current status.	VARCHAR(10)		Y	Allowed values: 'Available', 'Rented', 'Maintenance', 'Inactive'.
Insurance	Ins_Policy_No	Unique insurance policy number.	VARCHAR(20)	PK	Y	Unique alphanumeric identifier.
	Insurance_Name	Name of the insurance company/type.	VARCHAR(50)		Y	Alphabetic text.
	Ins_Begin_Date	Insurance start date.	DATE		Y	Must be a valid date.
	Ins_Expiry_Date	Insurance expiry date.	DATE		Y	Must be a valid date.
	Price_Per_Year	Annual insurance cost.	SMALLINT		Y	Range: 0 – 32,767
Reservation	Reservation_ID	Unique reservation identifier.	INT	PK	Y	Range: 1 – 2,147,483,647

	Customer_ID	Foreign Key to the reserving customer.	INT	FK (Customer)	Y	Must reference a valid Customer(Customer_ID) record.
	B_QR_Code	Foreign Key to the rented bicycle.	VARCHAR(10)	FK (Bicycle)	Y	Must reference a valid Bicycle(B_QR_Code) record.
	Start_Time	Rental start date and time.	DATETIME		Y	Must be a valid date and time.
	End_Time	Rental end date and time.	DATETIME		Y	Must be $\geq$ Start_Time.
	Payment	Total amount paid.	DECIMAL(7,2)		Y	Range: 0.00 – 99999.99
	Time_Lag	Duration of the rental (in minutes).	SMALLINT		Y	Range: 1 – 32,767 minutes.
	Payment_Method	Method of payment.	VARCHAR(15)		Y	Allowed values: 'Credit Card', 'Cash', 'Mobile Pay'.
Penalty	Penalty_ID	Unique penalty record identifier.	SMALLINT	PK	Y	Range: 1 – 32,767
	Reservation_ID	Foreign Key to the related reservation.	INT	FK (Reservation)	Y	Must reference a valid Reservation(Reservation_ID) record.
	Reason	Reason for the penalty.	VARCHAR(150)		Y	Descriptive text.
	Amount	Penalty amount.	SMALLINT		Y	Range: 0 – 32,767
Maintain_Personal	Staff_ID	Unique staff member identifier.	TINYINT	PK	Y	Range: 1 – 255
	Staff_Name	Name of the staff member.	VARCHAR(100)		Y	Alphabetic characters.
	Phone	Staff phone number.	VARCHAR(13)		Y	Mandatory phone number.
	E_Mail	Staff email address.	VARCHAR(100)			Optional email address.

	Role	Staff role/job title.	VARCHAR(20)		Y	Allowed values: 'Technician', 'Manager', 'Staff'.
Maintain	Maintain_ID	Unique maintenance record identifier.	SMALLINT	PK	Y	Range: 1 – 32,767
	B_QR_Code	Foreign Key to the bicycle under maintenance.	VARCHAR(10)	FK (Bicycle)	Y	Must reference a valid Bicycle(B_QR_Code) record.
	Staff_ID	Foreign Key to the staff member.	TINYINT	FK (Maintain_Personal)	Y	Must reference a valid Maintain_Personal (Staff_ID) record.
	Work_Description	Description of the maintenance work.	VARCHAR(255)		Y	Descriptive text.
	Cost	Maintenance cost.	SMALLINT		Y	Range: 0 – 32,767
	Next_Maintenance_Date	Scheduled next maintenance date.	DATE		Y	Must be a valid date ≥ today.
	Maintain_Date	maintenance date.	DATE		Y	Must be a valid date ≥ today.



Entity	Relationship	Connectivity	Entity
Bicycle_Class	Includes	1-1	Bicycle_Type
Bicycle_Class	Has	1-*	Bicycle_Model
Bicycle_Model	Defines	1-*	Bicycle
Bicycle	Needs	1-*	Maintain
Maintain_Personel	Performs	1-*	Maintain
Bicycle	Is Used In	1-*	Reservation
Reservation	Generates	1-1	Penalty
Insurance	Covers	1-*	Bicycle
Customer	Makes	1-*	Reservation